

VENDOR PERFORMANCE ANALYSIS REPORT

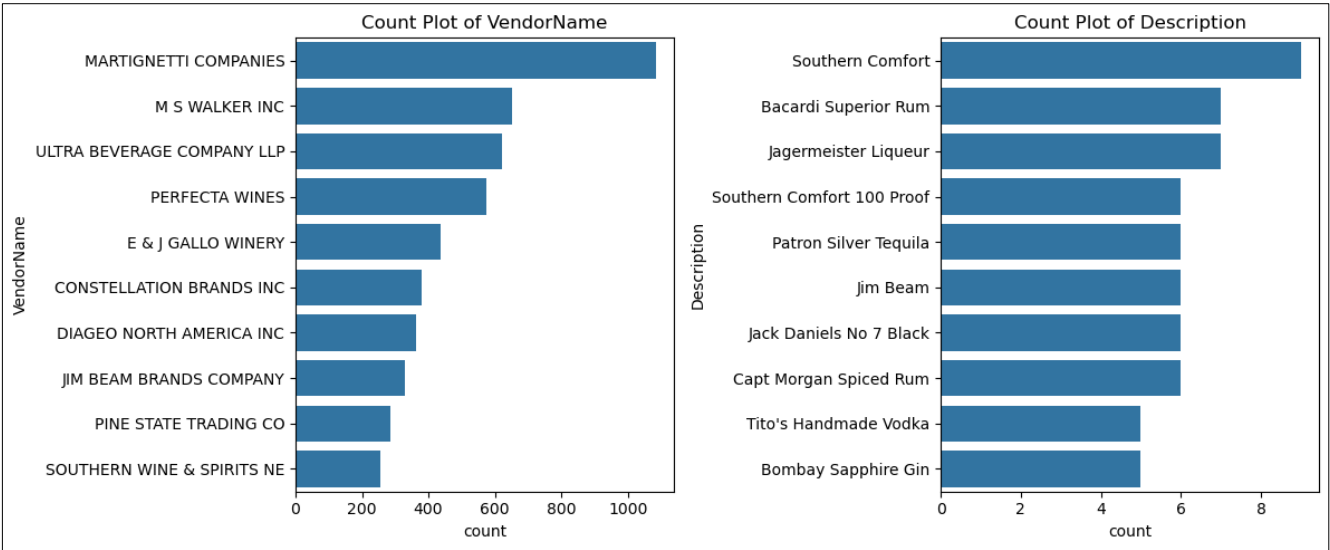
1. Problem statement:

Retail businesses operating across multiple stores and vendor relationships often struggle with visibility into vendor performance, pricing efficiency, and inventory health. Without a structured analysis, issues such as supplier concentration risk, loss-making transactions, slow-moving inventory, and inconsistent profit margins go undetected - leading to suboptimal procurement decisions and reduced profitability.

2. Objective:

The aim of this project is to conduct a comprehensive vendor performance analysis across 79 stores, 126 vendors, and 67 operational cities — identifying key patterns in procurement, sales, and inventory data. The analysis seeks to uncover inefficiencies in pricing, freight costs, and stock turnover, while providing actionable insights to support better vendor management, improve profit margins, and optimize overall business performance.

3. Vendor and product distribution analysis:



3.1. Key findings:

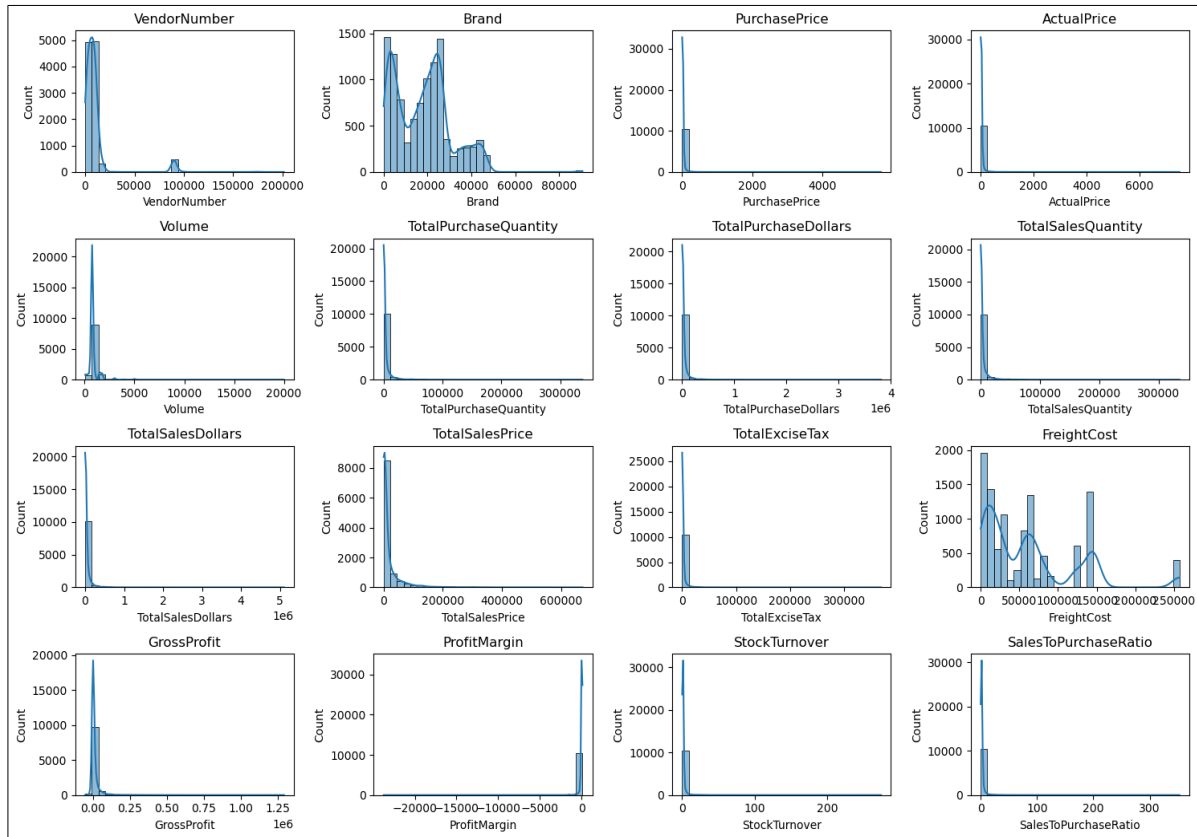
- 3.1.1. Procurement activities are concentrated among a small number of key suppliers, with MARTIGNETTI COMPANIES representing the largest share of vendor records.
- 3.1.2. In contrast, product representation is more evenly distributed across multiple beverage brands, indicating a diversified product portfolio.

3.2. Conclusion:

- 3.2.1. While product diversification reduces dependence on individual brands, the concentration of purchases among a limited number of suppliers may expose the business to supplier-related risks such as pricing pressure, supply disruptions, or contract dependency. These findings justify further vendor-level performance and procurement concentration analysis.

4. Exploratory data analysis:

	count	mean	std	min	25%	50%	75%	max
VendorNumber	10692.0	10650.649458	18753.519148	2.000000	3951.000000	7153.000000	9552.000000	2.013590e+05
Brand	10692.0	18039.228769	12662.187074	58.000000	5793.500000	18761.500000	25514.250000	9.063100e+04
PurchasePrice	10692.0	24.385303	109.269375	0.360000	6.840000	10.455000	19.482500	5.681810e+03
ActualPrice	10692.0	35.643671	148.246016	0.490000	10.990000	15.990000	28.990000	7.499990e+03
Volume	10692.0	847.360550	664.309212	50.000000	750.000000	750.000000	750.000000	2.000000e+04
TotalPurchaseQuantity	10692.0	3140.886831	11095.086769	1.000000	36.000000	262.000000	1975.750000	3.376600e+05
TotalPurchaseDollars	10692.0	30106.693372	123067.799627	0.710000	453.457500	3655.465000	20738.245000	3.811252e+06
TotalSalesQuantity	10692.0	3077.482136	10952.851391	0.000000	33.000000	261.000000	1929.250000	3.349390e+05
TotalSalesDollars	10692.0	42239.074419	167655.265984	0.000000	729.220000	5298.045000	28396.915000	5.101920e+06
TotalSalesPrice	10692.0	18793.783627	44952.773386	0.000000	289.710000	2857.800000	16059.562500	6.728193e+05
TotalExciseTax	10692.0	1774.226259	10975.582240	0.000000	4.800000	46.570000	418.650000	3.682428e+05
FreightCost	10692.0	61433.763214	60938.458032	0.090000	14069.870000	50293.620000	79528.990000	2.570321e+05
GrossProfit	10692.0	12132.381048	46224.337964	-52002.780000	52.920000	1399.640000	8660.200000	1.290668e+06
ProfitMargin	10692.0	-15.620770	443.555329	-23730.638953	13.324515	30.405457	39.956135	9.971666e+01
StockTurnover	10692.0	1.706793	6.020460	0.000000	0.807229	0.981529	1.039342	2.745000e+02
SalesToPurchaseRatio	10692.0	2.504390	8.459067	0.000000	1.153729	1.436894	1.665449	3.529286e+02

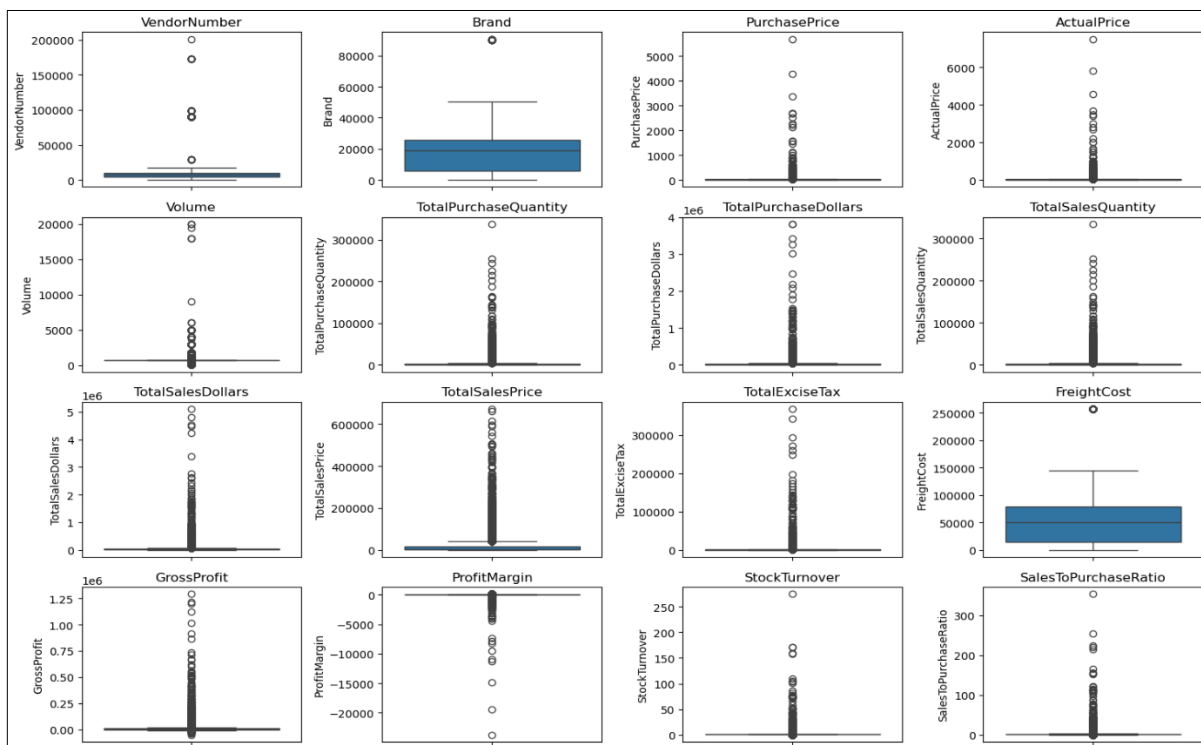


4.1. Negative & Zero Values:

- 4.1.1. Gross Profit: Minimum value is -52,002.78, indicating losses. Some products or transactions may be selling at a loss due to high costs or selling at discounts lower than the purchase price.
- 4.1.2. Profit Margin: Has a minimum of $-\infty$, which suggests cases where revenue is zero or even lower than costs.
- 4.1.3. Total Sales Quantity & Sales Dollars: Minimum values are 0, meaning some products were purchased but never sold. These could be slow-moving or obsolete stock.

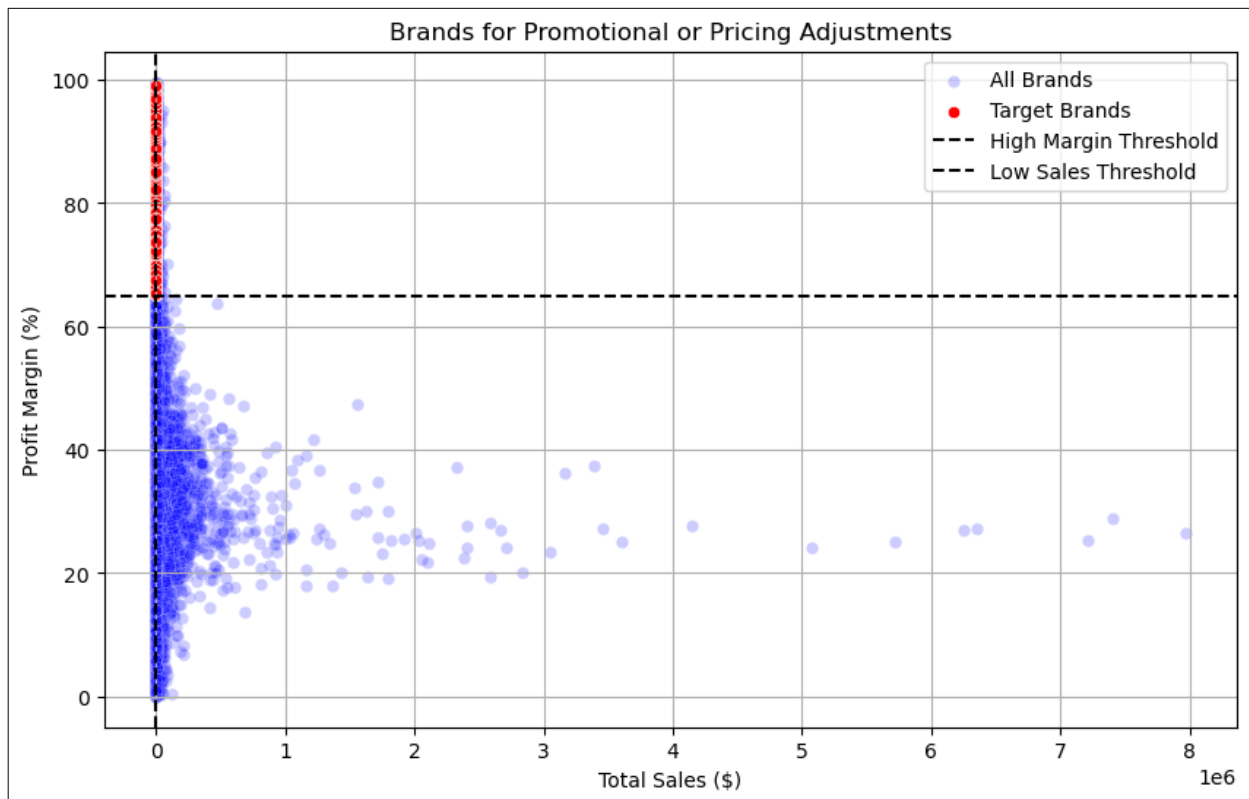
4.2. Outliers Indicated by High Standard Deviations:

- 4.2.1. Purchase & Actual Prices: The max values (5,681.81 & 7,499.99) are significantly higher than the mean (24.39 & 35.64), indicating potential premium products.
- 4.2.2. Freight Cost: Huge variation, from 0.09 to 257,032.07, suggests logistics inefficiencies or bulk shipments.
- 4.2.3. Stock Turnover: Ranges from 0 to 274.5, implying some products sell extremely fast while others remain in stock indefinitely. Value more than 1 indicates that sold quantity for that product is higher than purchased quantity due to either sale being fulfilled from older stock.

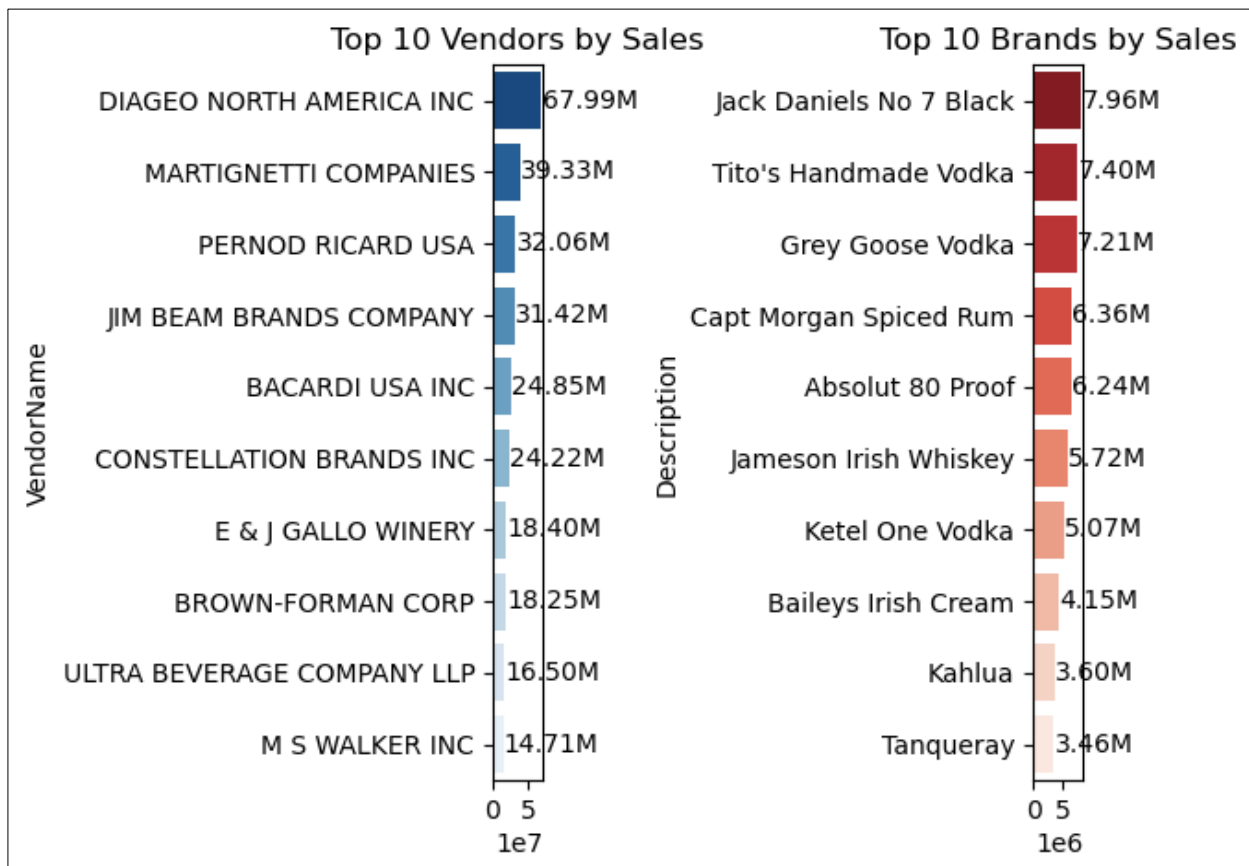


- 4.2.4. The boxplot analysis revealed substantial variability and the presence of numerous outliers across key financial and operational metrics. Revenue, purchase value, and gross profit distributions were heavily right-skewed, indicating that a small number of vendors and products contribute disproportionately to overall business performance.
- 4.2.5. Negative profit margin outliers suggest the existence of loss-making transactions that may require further investigation.
- 4.2.6. Additionally, inventory turnover and sales-to-purchase ratios displayed significant variation, highlighting differences in inventory efficiency across vendors and brands. These findings justify deeper vendor-level and brand-level analysis to identify opportunities for profitability improvement and inventory optimization.

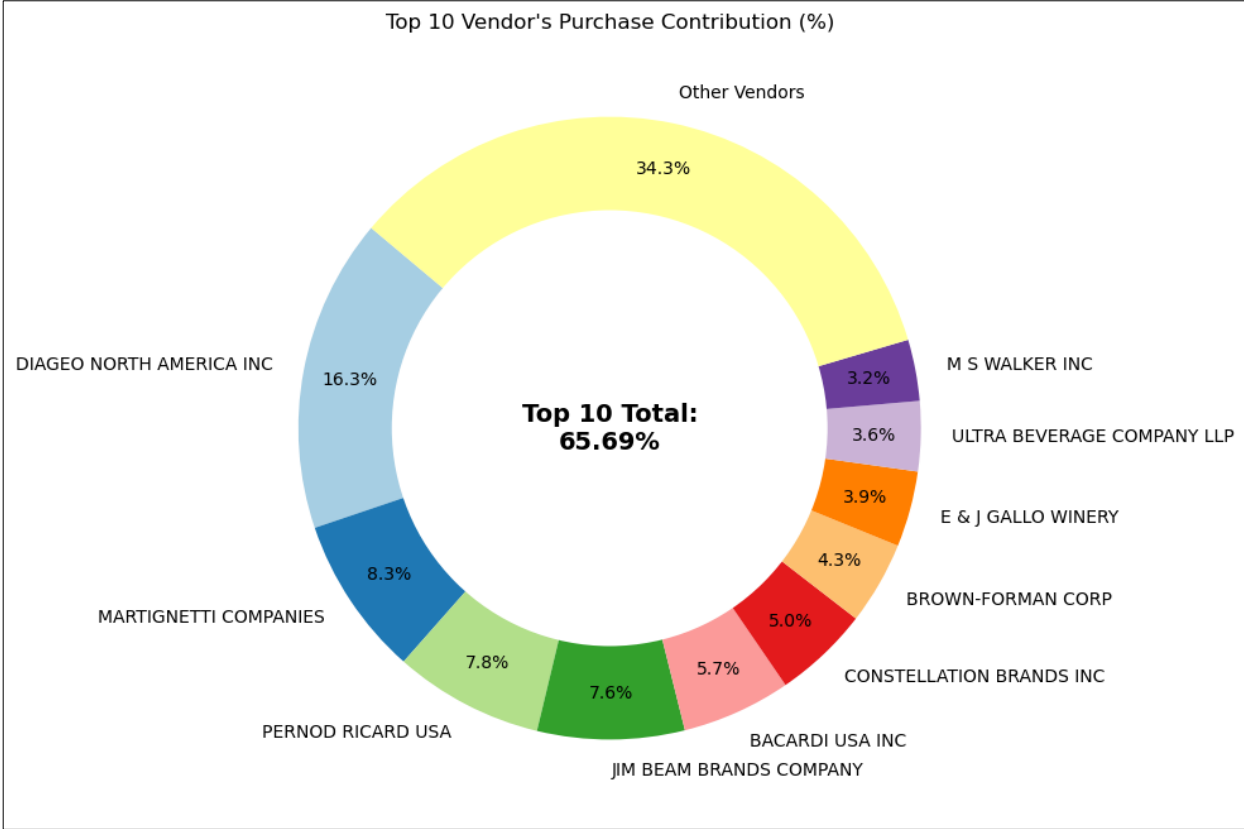
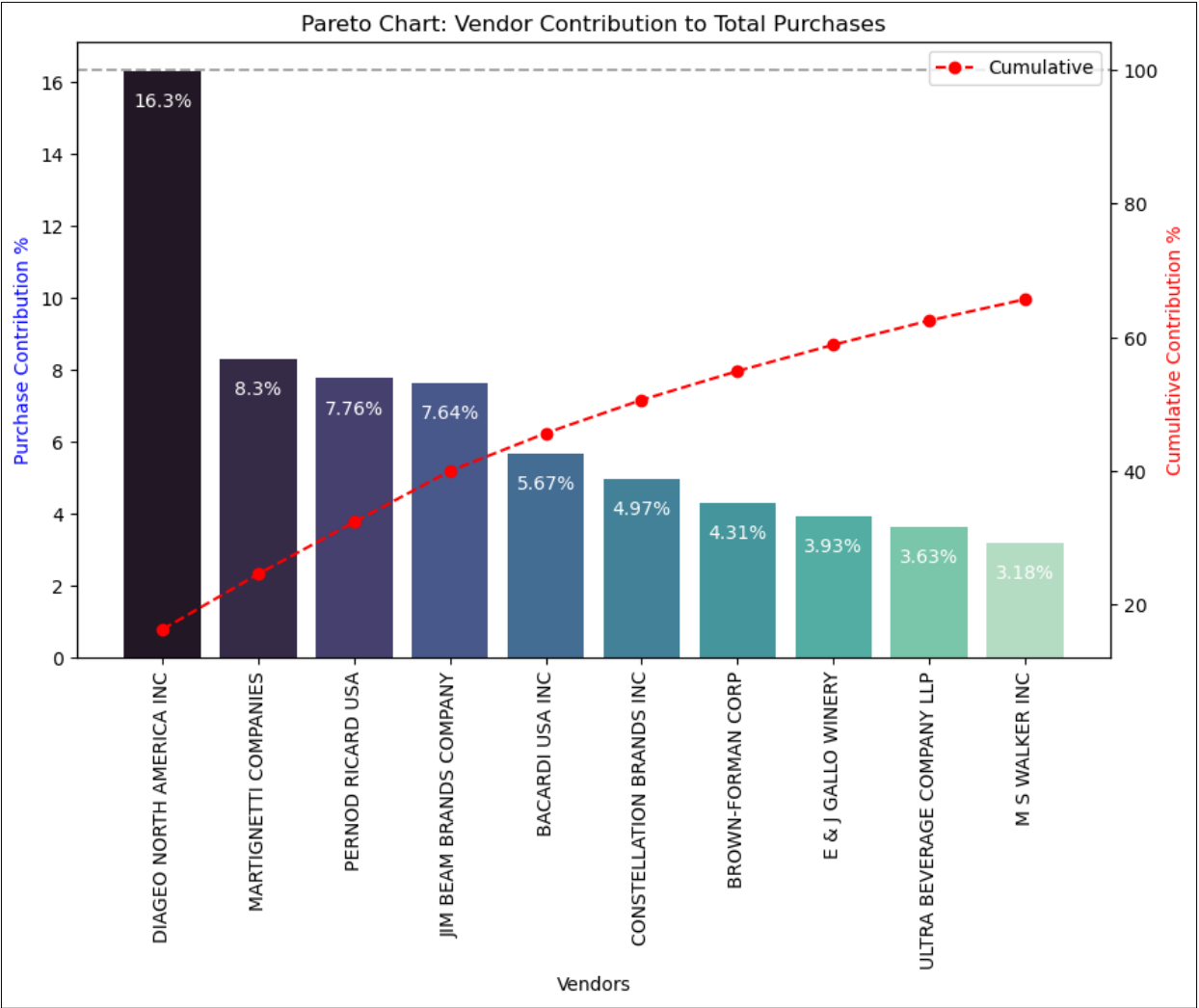
4.3. Brands that need Promotional or Pricing Adjustments which exhibit lower sales performance but higher profit margins.



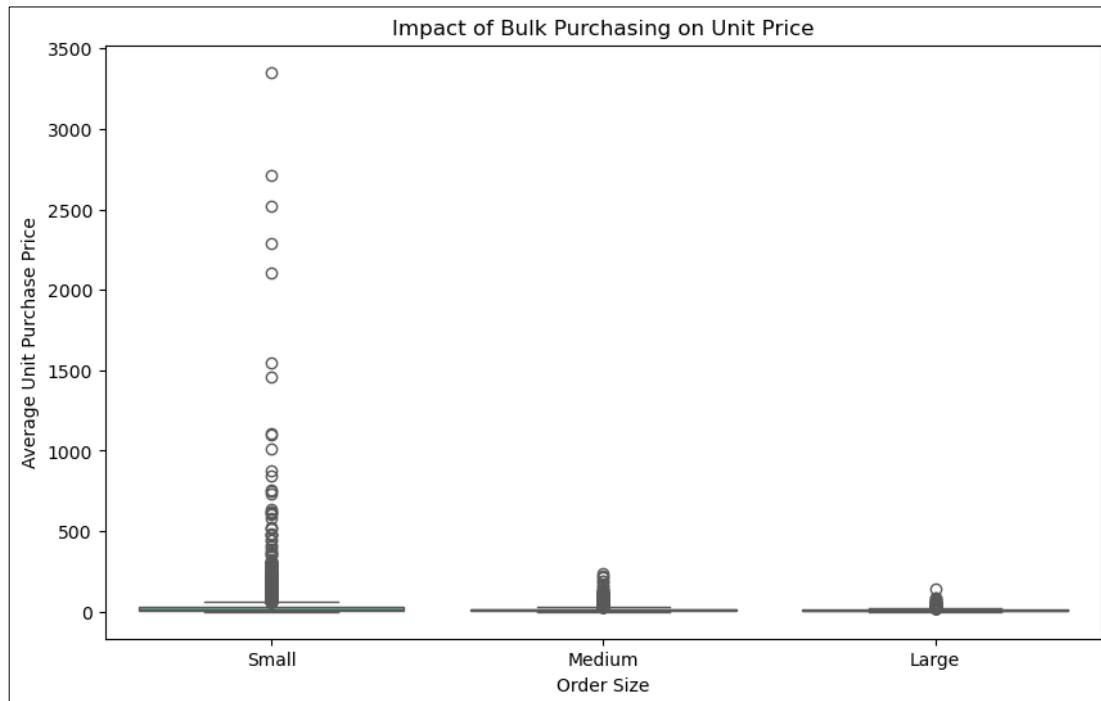
4.4. Vendors and Brands which demonstrates the highest sales performance:



4.5. Vendors contribute most to the total purchase:



4.6. Impact of purchase volume on unit purchase price:



4.6.1. Vendors buying in bulk (Large Order Size) get the lowest unit price (\$10.78 per unit), meaning higher margins if they can manage inventory efficiently.

4.6.2. The price difference between Small and Large orders is substantial (~72% reduction in unit cost).

4.6.3. This suggests that bulk pricing strategies successfully encourage vendors to purchase in larger volumes, leading to higher overall sales despite lower per-unit revenue.

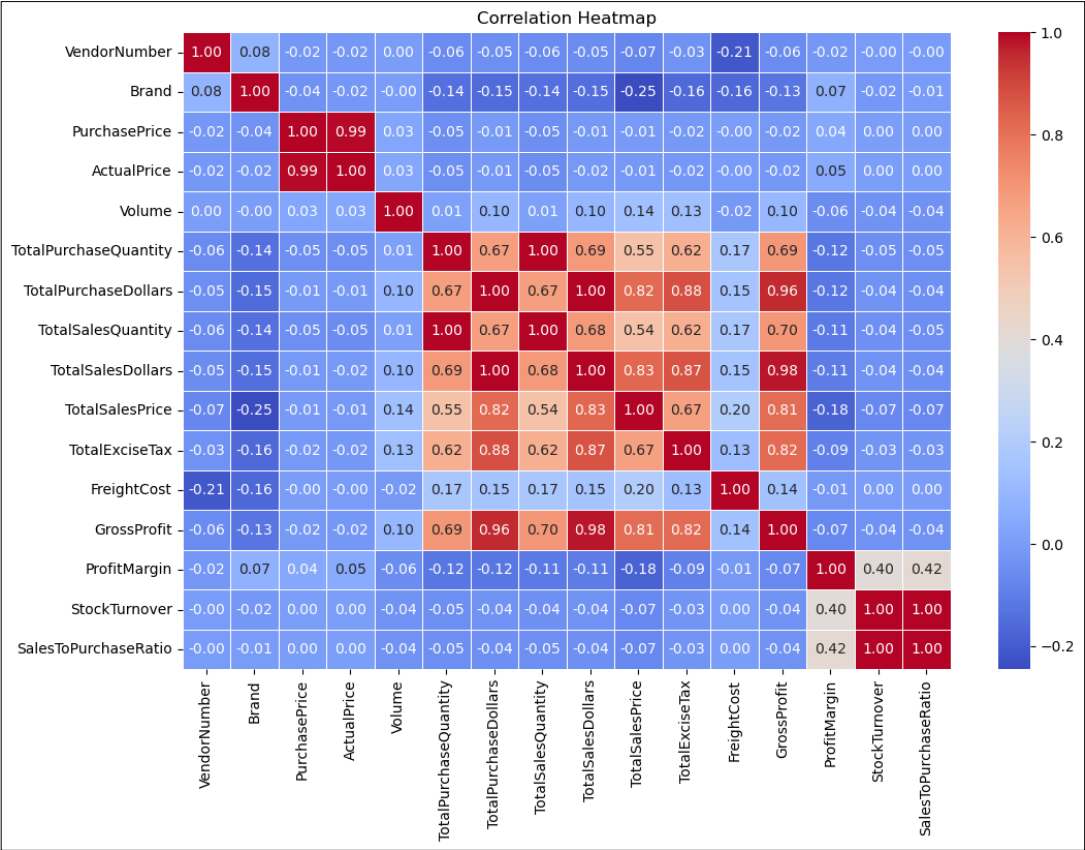
4.7. Vendors having low inventory turnover, indicating excess stock and slow-moving products:

StockTurnover	
VendorName	
ALISA CARR BEVERAGES	0.615385
HIGHLAND WINE MERCHANTS LLC	0.708333
PARK STREET IMPORTS LLC	0.751306
Circa Wines	0.755676
Dunn Wine Brokers	0.766022
CENTEUR IMPORTS LLC	0.773953
SMOKY QUARTZ DISTILLERY LLC	0.783835
TAMWORTH DISTILLING	0.797078
THE IMPORTED GRAPE LLC	0.807569
WALPOLE MTN VIEW WINERY	0.820548

4.8. How much capital is locked in unsold inventory per vendor and which vendor contributes the most to it?

	VendorName	UnsoldInventoryValue
25	DIAGEO NORTH AMERICA INC	722.21K
46	JIM BEAM BRANDS COMPANY	554.67K
68	PERNOD RICARD USA	470.63K
116	WILLIAM GRANT & SONS INC	401.96K
30	E & J GALLO WINERY	228.28K
79	SAZERAC CO INC	198.44K
11	BROWN-FORMAN CORP	177.73K
20	CONSTELLATION BRANDS INC	133.62K
61	MOET HENNESSY USA INC	126.48K
77	REMY COINTREAU USA INC	118.60K

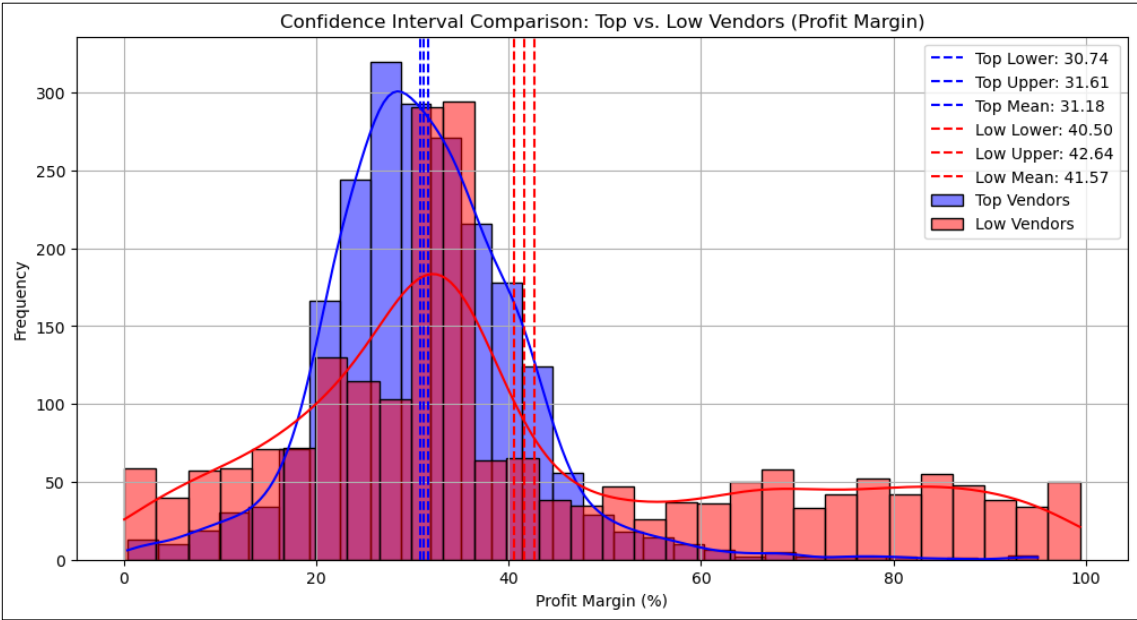
5. Correlation analysis:



- 5.1. Purchase Price has weak correlations with Total Sales Dollars (-0.012) and Gross Profit (-0.016), suggesting that price variations do not significantly impact sales revenue or profit.
- 5.2. Strong correlation between total purchase quantity and total sales quantity (0.999), confirming efficient inventory turnover.
- 5.3. Negative correlation between profit margin & total sales price (-0.179) suggests that as sales price increases, margins decrease, possibly due to competitive pricing pressures.
- 5.4. Stock Turnover has weak negative correlations with both Gross Profit (-0.038) and Profit Margin (-0.055), indicating that faster turnover does not necessarily result in higher profitability.

6. Confidence interval analysis and interpretation:

- 6.1. The confidence interval for low-performing vendors (40.48% to 42.62%) is significantly higher than that of top-performing vendors (30.74% to 31.61%).
- 6.2. This suggests that vendors with lower sales tend to maintain higher profit margins, potentially due to premium pricing or lower operational costs.
- 6.3. For High-Performing Vendors: If they aim to improve profitability, they could explore selective price adjustments, cost optimization, or bundling strategies.
- 6.4. For Low-Performing Vendors: Despite higher margins, their low sales volume might indicate a need for better marketing, competitive pricing, or improved distribution strategies



7. Suggestion for Business improvising:

- 7.1. Reduce Supplier Concentration Risk: Procurement is heavily concentrated among a few key suppliers (e.g., MARTIGNETTI COMPANIES) (Reference 3.1.1). Diversifying the supplier base will reduce dependency and minimize risks from pricing pressure, supply disruptions, or contract issues.
- 7.2. Address Loss-Making Transactions: Negative gross profit values (as low as -₹52,002.78) and infinite negative profit margins indicate certain products are being sold below cost (Reference 4.1.1). A pricing audit should be conducted to identify and correct these transactions.
- 7.3. Clear Slow-Moving and Obsolete Inventory: Some products have zero sales quantities despite being purchased (Reference 4.1.3). Introducing clearance strategies, promotions, or return-to-vendor policies can free up tied-up capital and reduce storage costs.
- 7.4. Optimize Freight and Logistics Costs: Freight costs vary enormously (from \$0.09 to \$257,032.07), signalling logistics inefficiencies (Reference 4.2.2). Consolidating shipments, renegotiating freight contracts, and improving delivery planning can bring these costs down.
- 7.5. Leverage Bulk Purchasing: Bulk orders see up to a ~72% reduction in unit cost (Reference 4.6). Encouraging vendors and categories to shift toward larger order sizes — where inventory can be managed efficiently — can significantly improve margins.
- 7.6. Boost Sales for High-Margin, Low-Volume Brands: Brands with high profit margins but low sales volume needs targeted promotional strategies, better distribution reach, or competitive pricing adjustments to convert their margin potential into actual revenue (Reference 4.3).
- 7.7. Improve Pricing Strategy for High-Performing Vendors: The negative correlation between sales price and profit margin suggests competitive pricing pressure. High-performing vendors should explore bundling, selective price adjustments, or value-added services to protect margins while sustaining volume (Reference 5.3).
- 7.8. Improve Inventory Turnover for Underperforming Vendors: Vendors with low stock turnover are locking capital in unsold inventory. Setting turnover targets, improving demand forecasting, and negotiating just-in-time delivery terms can help (Reference 4.7 & 4.8).
- 7.9. Investigate and Reduce Loss-Making Vendor Relationships: Some vendors contribute disproportionately to capital locked in unsold stock. A vendor performance review process should be established to renegotiate terms or phase out consistently underperforming vendor relationships.
- 7.10. Use Margin Insights to Guide Vendor Negotiations: Low-performing vendors have higher profit margins (40–43%) compared to top performers (30–32%) (Reference 6.1).. This insight can be used in negotiations — identifying where pricing or volume concessions can be made to grow sales without sacrificing profitability.

